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MILITARY SPECIFICATION

SYNTHETIC RUBBER COMPOUND, BUTADIENE-STYRENE TYPE, OZONE RESISTANT, FOR LOW TEMPERATURE SERVICE

1. SCOPE

1.1 Scope. - This specification covers requirements for a butadiene-styrene type molded rubber compound for serviceability at temperatures down to -65 degrees Fahrenheit.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards or such portions thereof as designated herein of the issue in effect on the date of invitation for bids form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-I-8660	Insulating and Sealing Compound, Electrical

STANDARDS

FEDERAL

Federal Test Method Standard No. 601	Rubber; Sampling and Testing

FSC 9320

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(Copies of specifications and standards, requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY TESTING METHODS

ASTM Designation: D 15-55T	Methods of sample preparation for physical testing of rubber products
ASTM Designation: D 518-44	Resistance to light checking and cracking of rubber compounds
ASTM Designation: D 1149-55T	Accelerated ozone cracking of vulcanized rubber

(Copies of ASTM Standards may be obtained from the American Society for Test Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Preproduction samples. - Unless otherwise specified in the contract or order (see 6.2), preproduction samples of the desired synthetic rubber compound of the same formulation and manufactured by the same methods and procedures as proposed for production under contract will be furnished as required. These samples will be sampled and tested as specified in 4 herein, and are for the purpose of determining that the product prior to production meets the requirements of end item use. Such samples will be retained with any samples subsequently submitted from production lots.

3.2 Materials. - The rubber material shall be a butadiene-styrene type combined with compounding ingredients, as may be required, of such nature and quality as to produce vulcanized stock conforming to this specification. A formulation and processing procedure found to be satisfactory for the desired compound is described in 6.3.

3.3 Color. - Unless otherwise specified, the color for the vulcanized stock shall be black.

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3.4 Physical property values. - Standard specimens of the compound shall conform to the property values shown in Tables I and II.

Table I - Physical property values for preproduction tests of snythetic rubber compound, butadiene-styrene type, ozone resistant, for low temperature service, when determined as prescribed in 4.6.

PROPERTIES	VALUES
TENSILE STRENGTH	
Initial	1475 psi (min)
After oven aging (70 hours at 212°F)	±15% change (max)
ELONGATION	
Initial	400% (min)
After oven aging (70 hours at 212°F)	-20% change (max)
HARDNESS (Shore "A" Durometer)	
Initial	55 ± 3
After oven aging (70 hours at 212°F)	60 (max)
After water immersion (70 hours at 212°F)	60 (max)
Low temperature (after 4 hours at -65°F)	75 (max)
TEAR RESISTANCE	
Initial	150 lbs/inch of thickness (min)
COMPRESSION SET	
After oven aging (22 hours at 158°F)	15% change (max)
After oven aging (70 hours at 212°F)	20% change (max)
STIFFNESS	
After 0.5 hours at -65°F	40° twist (min)
BRITTLENESS	
After 4 hours at -65°F	No failures
RESISTANCE TO SILICONE GREASE	
Hardness - Shore "A" Durometer (after treatment with silicone grease	60 (max)
Thickness (after treatment with silicone grease)	-5% change (max)
Weight (after treatment with silicone grease)	-10% change (max)

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WEIGHT CHANGE	
After oven aging (70 hours at 212°F)	-5% change (max)
VOLUME CHANGE	
After water immersion (70 hours at 212°F)	+5% change (max)
CONDUCTANCE	
Water extract	70 ohm $^{-1}$ cm $^{-1}$ x 10^{-6} (max)
SPECIFIC GRAVITY	1.15 ± 0.05
OZONE RESISTANCE	
After 7 days in 50 pphm ozone at 100°F	Crack free under 20X magnifications

Table II - Physical property values for production batch-acceptance tests of synthetic rubber compound, butadiene-styrene type, ozone resistant, for low temperature service, when determined as prescribed in 4.6.

PROPERTIES	VALUES
TENSILE STRENGTH (Initial)	1475 psi (min)
ELONGATION (Initial)	400% (min)
HARDNESS (Initial - Shore "A" Durometer)	55± 3

3.5 Workmanship. - The compound shall be manufactured in such a manner or such process as to assure compliance with all the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Performance of sampling inspection and testing. - Unless otherwise specified, sampling, inspection and testing for acceptance of individual lots shall be performed by the contractor subject to Government surveillance. Contractors not having laboratory facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the Government inspector. Detail records of all contractor performed examination and tests shall be made available to the Government inspector.

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4.2 Lot. - Unless otherwise specified, for the purpose of sampling, inspection and tests, a lot of the required rubber composition shall be limited to one manufacturers batch from a single rubber milling operation and shall not exceed 100 pounds.

4.3 Sampling. - Unless otherwise stated in the contract or order all units for test purposes shall be provided at the expense of the contractor.

4.3.1 Preproduction samples. - After award of contract but prior to entering quantity production, preproduction test specimen samples conforming to the requirements of 4.3.1.1, shall be delivered by the contractor to the laboratory designated by the procuring agency. (Any production started before approval shall be at the contractor's risk). These samples shall be manufactured by the same procedure and process and at the same location proposed by the contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended the contractor will be notified and will be authorized by the procuring agency to proceed with the submission of the product for acceptance.

4.3.1.1 Preproduction test specimen samples. - The manufacturer shall furnish the number of molded discs and panels shown in Table III.

Table III
Molded discs and panels for preproduction tests

Description	Number
Molded discs: 1.129 ± 0.005 inches diameter by 0.500 ± 0.005 inch thick	8
Molded panels 8 by 8 by 0.070 ± 0.005 inches	8

4.3.2 Preproduction samples for subsequent contracts. - The necessity for a preproduction sample will be determined by the procuring agency when production under a new contract by the same contractor at the same location follows the manufacturer's satisfactory production of material as covered by this specification.

4.4 Inspection.

4.4.1 Place of inspection. - Unless otherwise specified, inspection of material and equipment shall be at the point of the compound manufacture.

4.4.2 Visual inspection. - All samples taken for testing as specified in 4.5, shall be visually inspected to determine compliance with the material, workmanship and color requirements of this specification.

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4.5 Classification of tests. - The methods of tests shall be classified as follows:

a. Preproduction tests (see 4.5.1)
b. Production batch-acceptance test (see 4.5.2)

4.5.1 Preproduction tests. - These tests are specified in Table I, and are conducted on standard discs and panels conforming to the preproduction test specimen requirements of 4.3.1, on each compound for which preproduction evaluation is required.

4.5.2 Production batch-acceptance tests. - Batch-acceptance tests shall be performed in accordance with Table II, on test specimens which have been die stamped from a cured rubber panel 8 by 8 by 0.070 ± 0.005 inches from lots defined in 4.2. The cured rubber test panel shall be molded from the same lot as that used in making the finished part. Only lots or batches which meet the batch-acceptance test values specified in Table II of this specification shall be used in the manufacture of finished parts.

4.5.2.1 Test equipment and test facilities. - Unless otherwise specified by the procuring agency the manufacturer shall furnish and maintain all necessary facilities and equipment for making batch-acceptance tests. The test equipment shall be adequate in quantity to enable the testing to keep up with the production, and shall be of sufficient accuracy and quality to permit performance of the required tests.

4.5.2.2 Number of tests. - One set of test specimens as prescribed in 4.6.1.1, shall be considered sufficient for testing each batch. The average results for the specimens tested shall conform to the requirements prescribed in this specification.

4.5.2.3 Rejection. - Batch-acceptance lots shall be accepted or rejected by the inspector on the basis of the laboratory test report. Rejected lots may not be resubmitted.

4.5.2.4 Results of tests. - A copy of the results of the batch-acceptance tests shall be furnished the Government inspector.

4.6 Test procedures.

4.6.1 Standard test procedures.

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4.6.1.1 Number. - The minimum number of standard test specimens to be tested are specified in the test methods of 4.6.3.

4.6.1.2 Geometry. - The geometry of the standard test specimens shall be as specified in the applicable method of Federal Test Method Standard No. 601 and other specifications listed herein.

4.6.1.3 Molding of test discs and panels. - Test discs and panels shall be molded by methods which are representative of sound practice. No special treatment shall be used to improve the properties of the discs and panels when compared with articles molded in commercial production.

4.6.2 Test conditions. - The synthetic rubber compound shall be subjected to tests under the following conditions:

4.6.2.1 Ambient temperature and humidity. - Unless otherwise specified, tests listed under 4.6, except those listed in 4.6.2.2, shall be conducted as prescribed in Section 7 of Federal Standard FED-STD-601.

4.6.2.2 Low temperature. - The low temperature test for Hardness, Stiffness, and Brittleness as prescribed in 4.6.3.3.2, 4.6.3.6 and 4.6.3.11, respectively, shall be conducted at minus 65 ± 2°F.

4.6.2.3 Oven aging temperature. - The oven aging tests shall be subjected to either 158 ± 2°F (70 ± 1.1°C) or 212 ± 3°F (100 ± 1.7°C)

4.6.3 Test methods. - The test methods shall be in conformance with the applicable method of Federal Test Method Standard No. 601 and modified as follows:

4.6.3.1 Tensile strength.

4.6.3.1.1 Initial tensile strength. - Tensile strength at room temperature shall be determined by Method 4111 of Standard FED-STD-601 on dumbell specimens prepared in accordance with figure 4111, Die IV. Results of tests on specimens which break outside of the straight reduced section or which are found to be imperfect shall be discarded and retests shall be made. The test specimen shall be between 0.065 and 0.075 inches thick. Test results shall be calculated from an average of 5 specimens.

4.6.3.1.2 Tensile strength after aging. - After oven aging for 70 hours at 212°F, the tensile strength shall be determined as prescribed in 4.6.3.1.1. Determination of oven-aged properties shall be made not less than 16 hours nor more than 48 hours after removal from the oven. Tension tests on unaged specimens shall be made immediately prior to and on the same machine as the tension tests on the aged specimens. Test results shall be calculated from an average of 5 specimens.

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4.6.3.2 Elongation. - The ultimate elongation of test specimens shall be determined by Method 4121, of Standard FED-STD-601 on dumbell specimens prepared in accordance with Method 4111, Figure 4111, Die IV, of Standard FED-STD-601. Test results for each condition shall be calculated from an average of 5 specimens.

4.6.3.3 Hardness.

4.6.3.3.1 Initial hardness; after oven aging for 70 hours at 212°F; and after immersion in water for 70 hours at 212°F. - The Shore "A" durometer hardness after each of the above test conditions shall be determined as prescribed in Method 3021, of Standard FED-STD-601. Four test strips, 2 by 1 by 0.070 inches shall be plied up for hardness measurements. Three hardness measurements shall be taken on one face and three on the opposite face of the plied up specimen. Test results for each condition shall be expressed as the average of the six readings.

4.6.3.3.2 Low temperature. - The Shore "A" durometer hardness of test specimens after conditioning for four hours at -65°F, shall be determined as prescribed in Method 5521, of Standard FED-STD-601. Four test strips, 2 by 1 by 0.070 inches shall be plied up for hardness measurements. Three hardness measurements shall be taken on one face and three on the opposite face of the plied up specimen. Test results for each condition shall be expressed as the average of the six readings.

4.6.3.4 Tear resistance. - The tear resistance of the rubber specimen shall be determined as prescribed in Method 4211, of Standard FED-STD-601, on a test specimen prepared in accordance with Figure 4211, Die B. Test results shall be calculated from an average of 5 specimens.

4.6.3.5 Compression set.

4.6.3.5.1 After compression at 158°F. - The compression set of test specimens under compression for 22 hours at 158°F, shall be determined by Method 3311, of Standard FED-STD-601. The test specimen shall be a molded disc 1.129 inches in diameter and one-half inch thick. Test results shall be calculated from an average of 2 specimens.

4.6.3.5.2 After compression at 212°F. - The compression set of test specimens under compression for 70 hours at 212°F shall be determined as prescribed in 4.6.3.5.1. Test results shall be calculated from an average of 2 specimens.

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4.6.3.6 Stiffness. - The stiffness of the test specimens, measured in degrees of twist after 0.5 hours at -65°F, shall be determined as prescribed in Method 5611, of Standard FED-STD-601. The test specimen strip shall be 1.5 inches by 0.070 inches. The torsional constant of the wire shall be 0.125 gram-centimeters per degree of twist. The effective length of the rubber test strip shall be one inch. Test results shall be calculated from an average of 2 specimens.

4.6.3.7 Resistance to silicone grease. - Apply a coat of silicone grease conforming to Specification MIL-I-8660, approximately one-sixteenth inch thick on all faces of four 2 by 1 by 0.070 inch test specimens. Place the test specimens in a glass beaker and heat in an oven for 336 hours at 158°F. After the exposure period, remove the silicone grease from each specimen with a spatula and wipe thoroughly with paper toweling. Test for hardness, thickness and weight change as follows:

4.6.3.7.1 Hardness. - Determine hardness on plied up specimens as prescribed in Method 3021, of Standard FED-STD-601, using a Shore "A" durometer. Four test strips, 2 by 1 by 0.070 inches shall be plied up for hardness measurements. Three hardness measurements shall be taken on one face and three on the opposite face of the plied up specimen. Test results for each condition shall be expressed as the average of the six readings.

4.6.3.7.2 Thickness. - The thickness of test specimens shall be measured with a micrometer or dial gauge before and after the specimens have been conditioned as prescribed in 4.6.3.7. Measurements before and after conditioning shall be taken in the same spot. The decrease in thickness shall be recorded as percent change. Test results shall be calculated from an average of 3 specimens.

4.6.3.7.3 Weight change. - The weight of each test specimen shall be measured in grams on a balance accurate to ± 0.001 grams before and after the specimens have been conditioned as prescribed in 4.6.3.7. The weight decrease shall be recorded as percent change. Test results shall be calculated from an average of 3 specimens.

4.6.3.8 Weight change after oven aging for 70 hours at 212°F. - The percent decrease in weight after oven aging shall be determined on test specimens 2 by 1 by 0.070 inches. The specimens shall be held overnight in a desiccator and weighed to the nearest mg before subjecting to the oven aging. After removal from the oven, specimens shall be cooled in a desiccator for not less than 2 hours nor more than 70 hours and again weighed to the nearest mg. Test results shall be calculated from an average of 3 specimens.

4.6.3.9 Volume change after water immersion for 70 hours at 212°F. The percent increase in volume after water immersion for 70 hours at 212°F shall

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be determined as prescribed in Method 6211 of Standard FED-STD-601. Three test specimens, 2 by 1 by 0.070 inches, shall be immersed in 300 ml of distilled water. Test results shall be calculated from an average of three specimens.

4.6.3.10 Water extract conductance. - This method covers the determination of water soluble ionic constituents in molded rubber materials which contribute to the acceleration of galvanic corrosion of adjacent metal surfaces in confined humid atmospheres. It is also intended to give semi-quantitative information regarding the content of soluble acids, bases and salts in molded parts after exposure at 158 ± 2°F for 288 hours in direct contact with water. The specific conductance of the aqueous extract is measured and is taken as an index of the likelihood that in humid atmosphere adjacent metal surfaces might be corroded by galvanic action or by direct chemical attack. The test does not provide an absolute measure of the concentration of ionic substances in a material, but it does furnish evidence for predicting the potential corrosiveness of rubber materials.

4.6.3.10.1 Apparatus. - The apparatus required for the test will consist of equipment common to testing laboratories and the following items:

(a) Conductance bridge. - A conductance type Wheatstone bridge having a range of one to 250,000 ohms measured resistance, a built-in potentiometer, a 1000 ± 50 cycle per second oscillator, and a sensitive null point indicator. Resistance measurements must be accurate to at least ± 2%. An instrument as manufactured by Arthur H. Thomas Company, West Washington Square, Philadelphia 5, Pennsylvania, Catalog No. 3963 has been used satisfactorily.

(b) Conductivity cell. - A dipping type micro conductivity cell for solutions of high or medium conductance with a cell constant of approximately 1.0 cm^{-1} and with a maximum outside tube diameter of 0.5 inch. An instrument as manufactured by Arthur H. Thomas Company, West Washington Square, Philadelphia 5, Pennsylvania has been used satisfactorily.

4.6.3.10.2 Reagents.

4.6.3.10.2.1 Distilled water. - Water obtained from a copper still on a single distillation and stored at ambient temperatures in pyrex glass bottles is satisfactory so long as the measured specific conductance is less than 2.0 by 10^{-6} $ohm^{-1}cm^{-1}$.

4.6.3.10.2.2 Potassium chloride solution. - Dry reagent grade KCl at 105 ± 2°F for 24 hours. Cool to room temperature, weigh out exactly 0.7463 grams of KCl, and add to 1000 grams of distilled water. Shake

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thoroughly. Store in a glass stoppered pyrex reagent bottle. At 23.0 ± 0.5 °C (73.4° F) the specific conductance of this solution is 0.001355 ohm^{-1} cm^{-1} not including the conductance of the distilled water alone. This solution will be used only for the determination of the cell constant of the conductivity cell.

4.6.3.10.3 Test specimens. - Samples shall be obtained from 8 by 8 by 0.075 inch sheet molded according to ASTM D15-55T.

4.6.3.10.4 Procedure

(a) Sufficient material shall be cut from 8 by 8 by 0.070 inch sheet rubber into 0.070 inch average cubes for test samples.

(b) Weigh 0.50 ± 0.01 gram portions of the rubber cubes into 3 clean dry 50 milliliter Erlenmyer pyrex flasks. Seal with ground glass stoppers.

(c) When a number of flasks are ready for activation, pipet into each flask 50.0 milliliter of distilled water, stopper tightly and shake until the rubber particles are wetted.

(d) Mark on the wall of the flask with a glass scribe the exact level of the water in the flask.

(e) Store in an oven at 160 ± 6°F for 288 hours. At the end of the first day of oven storage shake each flask to dislodge adsorbed air bubbles that tend to float the rubber particles and prevent wetting of all exposed surfaces.

(f) Transfer from the oven to a constant temperature bath adjusted to 73.4° F (23.0° C). When thoroughly cooled add just enough distilled water to bring the water levels in each flask to the original volume mark. Shake thoroughly and allow the mixture to settle out.

(g) Determine the cell constant of the conductivity cell. Add 50.0 milliliters of 0.01000 demal standard potassium chloride solution to a clean dry 50 milliliter flask identical to the test flasks. Lower the dipping type conductivity cell vertically into the KCl solution until the bottom edge of the cell rests on the bottom of the flask. Bring assembly and solution to 23.0 ± 0.5°C (73.4 ± 1°F) by immersion in a constant temperature bath. Measure the specific resistance in ohms of the KCl solution at 1000 cycles a. c.

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(h) With the same technique and the same conductivity cell measure the specific resistance of each of the solutions extracted from the test rubbers.

4.6.3.10.5 Calculations

4.6.3.10.5.1 Cell constant. - The conductivity cell constant K is given by K = kR where k is the specific conductance of the standard KCl solution (0.001355 ohm $^{-1}$ cm $^{-1}$ for 0.01000 demal KCl at 23.0 ± 0.5°C and R is the observed resistance in ohms of the KCl solution in the cell at the same temperature. K should be approximately 1.0 cm^{-1}.

4.6.3.10.5.2 Conductance ratings of rubbers. - The average specific conductance in ohms $^{-1}$ cm $^{-1}$ of the aqueous extracts of three samples of each material after 288 hours storage at 158°F. will be called the conductance rating of the rubber and will be calculated from the relation

$$\text{Conductance rating of material} = \frac{3K}{R_1 + R_2 + R_3}$$

where K = the conductivity cell constant and R_1, R_2, R_3 are the observed resistances in ohms of the three extracts.

4.6.3.10.6 Accuracy

(a) Test results may be expected to vary within a given lot in the limits of the mean of three samples ± 5% for most materials. The percentage variation increases with increasing conductance rating.

(b) Lot to lot variation may approach ± 5% of the mean of five lots and will increase with increasing conductance rating.

4.5.3.10.7 Report

(a) Material designation
(b) Name of manfuacturer
(c) Date and time in oven at 160°F
(d) Date and time out of oven
(e) Hours storage time at 160°F
(f) Cell constant K of conductivity cell
(g) Resistance in ohms

Flask No.	Resistance (Ohms) 1000 c/s a.c.
1	
2	
3	
Sum: 1, 2, and 3 =	

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(h) Average specific conductance L_s in ohms $^{-1}cm^{-1} \times 10^{-6}$

$$L_s = 3K / R_1 + R_2 + R_3$$

4.6.3.10.8 Safe limits based on conductance rating

4.6.3.10.8.1 For preproduction acceptance. - After 288 hours of exposure at 158°F, the maximum average conductance rating for three samples shall be 70 by 10^{-6} ohm^{-1} cm^{-1}.

4.6.3.11 Low temperature brittleness. - This method is intended for use in determining the brittleness of vulcanized rubber compounds at low temperature by use of a "Brittleness Tester". (See Figure 1.)

4.6.3.11.1 Specimens. - Five test strips each, 1.5" by 0.125" by 0.070 inches, shall be prepared. The principal surfaces of the specimens shall be parallel to within 0.010 inches.

4.6.3.11.2 Apparatus

4.6.3.11.2.1 Brittleness tester. - The brittleness tester (Figure 1), shall consist of five (5) open and non-corrosive guide tubes, each measuring 1.125 inches on the inside diameter and approximately 18.5 inches in length. The guide tubes shall be positioned parallel to each other, vertically oriented and secured in place on a wood platform with corrosion resistant clamps and bracing. A hole, 0.312 inches in diameter, located 14.37 inches from the bottom of the guide tube shall be drilled on center through each of the five tubes. A 0.250 inch rod designated as the ball release rod is inserted through these holes. The purpose of the tube is to act as a guide for a falling ball. The distance from the top of the inserted ball release rod to the rubber test specimens shall be 14.56 ± 0.06 inches.

4.6.3.11.2.2 Specimen holder blocks and specimen support blocks. - Two wooden "specimen holder blocks" one inch by one inch by desired length position the test specimens under the guide tubes. A wooden "specimen support" block one inch by one inch by desired length supports the flexible free end of the test specimens during the cold conditioning process.

4.6.3.11.2.3 Test balls. - Five spherical, steel, test balls, each one inch in diameter and weighing 65 ± 1.0 grams are required.

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4.6.3.11.2.4 Low temperature cabinet. - A low temperature cabinet designed as prescribed in Method 5111, of Standard FED-STD-601, shall be used. The low temperature cabinet cooling chamber shall be large enough to contain the brittleness tester. A 0.375 inch hole shall be drilled in the side of the cabinet for the extension of the ball release rod to the outside of the cabinet.

4.6.3.11.3 Test procedure. - Position the lower specimen holder block and specimen support block as shown in Figure 1. Adjust specimen support block so that its height is level with the lower specimen holder block. Place a test specimen on the blocks perpendicular to and under each tube so that 0.5 inch of the specimen lays on the specimen holder block and 1.0 inch on the specimen support block. Place the upper specimen holder block on the lower specimen holder block and secure the specimen blocks to the platform with C-clamps (not shown in Figure 1). Position the brittleness tester in the low temperature cabinet in a manner that will align the hole in the cabinet with the ball release rod holes in the guide tubes. Insert the ball release rod through the cabinet and all the guide tubes. Bring the temperature within the cabinet to minus 65°F and hold for three (3) hours. After conditioning period, remove the specimen support block and insert one ball in each tube. Continue to condition at minus 65°F for another hour. Withdraw ball release rod. Remove the brittleness tester from cabinet and observe effects on rubber.

4.6.3.11.4 Results

4.6.3.11.4.1 Unless otherwise specified, five specimens from each lot shall be tested.

4.6.3.11.4.2 Upon visual inspection of the specimens, the absence of rupture, cracks or breaks shall indicate compliance with lot temperature brittleness requirements. Existence of breaks, cracks or rupture shall indicate failure and be considered as not conforming to the requirements listed under "Brittleness" in Table 1.

4.6.3.11.4.3 Rejection of the lot is mandatory if there are any failures.

4.6.3.12 Specific gravity. - The specific gravity of the vulcanized rubber compound shall be determined by Method 14011 of Standard FED-STD-601. The test specimens shall be strips 2 by 1 by 0.070 inches. Test results shall be calculated from an average of 3 specimens.

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4.6.3.13 Ozone resistance. - The ozone resistance of the vulcanized rubber compound shall be determined in conformance with the method described in ASTM D 1149-55T, and modified as follows:

4.6.3.13.1 Test specimens. - Two test strips each, 1.0 inch by 3.75 inch by 0.070 inches, shall be prepared and mounted in accordance with Method B (bent loops) of ASTM D 518-44.

4.6.3.13.2 Ozone concentration. - The ozone concentration shall be 50 ± 5 parts per 100,000,000 of air by volume. The ozone concentration shall be determined at least once each day.

4.6.3.13.3 Procedure. - Specimens shall be exposed at 100 ± 2°F for 168 hours in an atmosphere conforming to 4.6.3.13.2.

4.6.3.13.4 Results. - Inspection of the specimens under 20X magnifications shall be made immediately upon removal from the test chamber and prior to unmounting of the specimens. The absence of cracks on the bent loop portion of the specimen shall indicate compliance with ozone resistance requirements. Existence of cracks shall indicate failure and be considered as not conforming to the requirements listed under "ozone resistance" in Table I.

4.6.3.13.5 Report

(a) Material designation

(b) Name of manufacturer

(c) Date and time of mounting of specimens in holders

(d) Date and time in ozone cabinet at 100°F

(e) Date and time out of ozone cabinet

(f) Date, time, and results of daily ozone concentration determinations

(g) Results of inspection of specimens under 20X magnifications

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5. PREPARATION FOR DELIVERY OF TEST SPECIMENS

5.1 There are no preparation, packaging or packing requirements for shipping test specimens on rubber compounding ingredients. Shipments shall be in accordance with good commercial practice.

6. NOTES

6.1 Intended use. - This rubber compound is intended for use in the manufacture of diaphragms and boots that require flexing at minus 65°F.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Preproduction sampling waiver where applicable

6.3 Introduction to formulating and processing procedure. - The following sub-paragraphs describe an advisory procedure for formulating and processing a "Synthetic Rubber Compound, Butadiene-Styrene Type, Ozone Resistant, for Low Temperature Service", and outlined are the minimum standards of care considered essential to maintain acceptable performance levels in the desired vulcanized rubber product. The detailed description of the formulation and processing is for the purpose of emphasizing procedural points and is not intended to provide an inflexible process precluding improvements based on sound investigation. Therefore, any and all references to commercial types, styles, trade names, and catalogs are intended to be descriptive and not restrictive and only to indicate the characteristics of articles which will be satisfactory.

6.3.1 Formulation. - A formulation which has produced acceptable material is outlined in Table IV.

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TABLE IV
FORMULATION

ITEM NUMBER	COMPOUNDING INGREDIENTS	PARTS BY WEIGHT	SUPPLIER
1	Naugapol 1023	100 ± 5	Naugatuck Chemical Co. Naugatuck, Conn.
2	Philblack 0	40 ± 2	Phillips, Chemical Co. Akron 8, Ohio
3	HI SIL 233	20 ± 1	Columbia-Southern Chem. Corp., Pittsburg, Pa.
4	TP 90B	15 ± 0.75	Thiokel Corp. Trenton, N. J.
5	Zinc oxide (Kadox 25)	3 ± 0.15	New Jersey Zinc Sales Co. New York 38, N. Y.
6	Stearic acid N. F.	1.5 ± .08	A. T. Baker Chemical Co. Phillipsburg, N. J.
7	Heliozone	1 ± 0.05	E. I. Dupont, Rubber Chem. Division, Wilmington, Del.
8	UOP 88	5 ± 0.25	Universal Oil Products Des Plains, Ill.
9	Neozone - D	0.5 ± .05	E.I. Dupont, Rubber Chem. Div., Wilmington, Del.
10	Sulphur (Blackbird)	0.7 ± .05	C. P. Hall Co. Akron 8, Ohio
11	Monex	2 ± 0.1	Naugatuck Chemical Co. Naugatuck, Conn.
12	D. P. G.	0.4 ± .05	American Cyanimid Co. New York 20, N. Y.

6.3.2 Processing procedure. - The sequence of compounding ingredient additions and processing techniques are as follows:

6.3.2.1 Band and masticate the Naugapol 1023 on a water cooled mill.

6.3.2.2 Add item 2, gradually, and approximately 1/3 of item 4, as needed to maintain a band on the rolls.

6.3.2.3 Add item 3, gradually, and approximately 1/3 of item 4, as needed to maintain a band on the rolls.

6.3.2.4 Add all of item 5 and the remainder of item 4.

6.3.2.5 Add ingredients 6 through 12 in order and mill at least five minutes between additions.

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6.3.2.6 After the addition of all materials, mill for 15 minutes.

6.3.2.7 Pass the rubber through tight rolls 20 times. Ball up the rubber prior to each pass through the rolls.

6.3.2.8 On the last pass of the rubber through the mill, open the rolls to get the desired thickness.

6.3.2.9 Processing techniques.

6.3.2.9.1 Loosen rolls with gradual addition of ingredients, so as to maintain approximately a one inch bank above the rolls during steps 6.3.2.2, through 6.3.2.5.

6.3.2.9.2 Cut rubber on rolls as necessary, to maintain movement in the bank and to aid in the incorporation of ingredients items 2, 3, 5, 6, 7, 9, 10, 11 and 12.

6.3.2.9.3 Cut rubber frequently on the rolls, using a diagonal cut. Draw cut end out and fold into bank. Alternate the diagonal cuts from each side during steps 6.3.2.2, through 6.3.2.6, so as to achieve uniform blending.

6.3.2.9.4 The time required to mill a two pound batch on a 12 inch laboratory mill is approximately 1.5 hours. Inadequate milling may produce a non-homogeneous rubber compound that will fail to meet strength requirements.

6.3.3 Cure of test panels. - The curing of test panels measuring 8 by 8 by 0.070 inches shall be for 30 minutes at 300 ± 5° Fahrenheit.

6.3.4 Cleanliness. - Materials and processing must be clean to insure that the final product will be free of foreign particles.

Patent notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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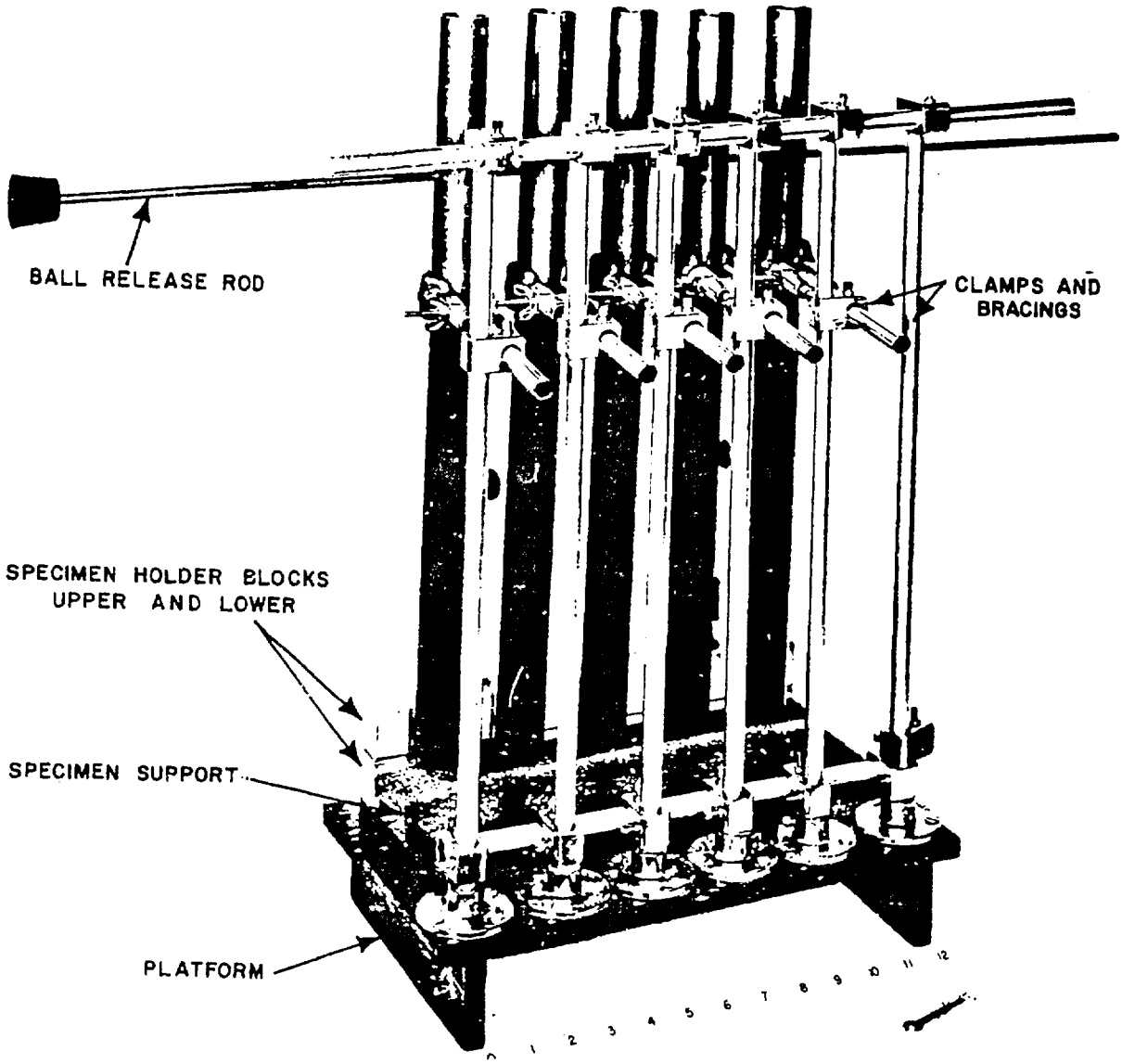


FIG. 1 BRITTLENESS TESTER

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